Serialization and networking

Code generation, equality, HTTP, retry

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Lab objectives

Generate JSON code

Writes fromJson and toJson for you.

Compare by value

Same fields make two todos equal.

Fetch live data

dio loads the list from a public API.

Retry with backoff

Delay and jitter survive slow servers.

Time
2 hours

Present
To the lab instructor, at the end of this lab

File layout

1. Continue in the project from Lab 3.
2. Two new files: `lib/models/todo.dart` and `lib/api/todo_api.dart`.
3. Add the packages with the two commands below.
4. After each model change: `dart run build_runner build`

```
flutter pub add json_annotation equatable dio
flutter pub add dev:build_runner dev:json_serializable
```

Todo model: code generation and equality

1. Define `Todo` with `userId`, `id`, `title`, `completed`, all final.
2. Annotate the class with `@JsonSerializable()` and add the `part` directive.
3. Add typed `fromJson` and `toJson` factory methods.
4. Run build_runner and confirm `todo.g.dart` appears.
5. Extend `Equatable` and list every field in `props`.
6. Build two identical todos and print `a == b`.

Done when

- [ ] `todo.g.dart` is generated.
- [ ] `a == b` prints `true`.
- [ ] `flutter analyze` reports no issues.

Code generation for Todo

```dart
part 'todo.g.dart';
@JsonSerializable()
class Todo {
  final int userId, id;
  final String title; final bool completed;
  const Todo({required this.userId, required this.id,
      required this.title, required this.completed});
  factory Todo.fromJson(Map<String, dynamic> j) => _$TodoFromJson(j);
  Map<String, dynamic> toJson() => _$TodoToJson(this);
}
```

Run build_runner after every model change to regenerate `todo.g.dart`.

Value equality with Equatable

```
class Todo extends Equatable {
  final int userId, id;
  final String title;
  final bool completed;
  const Todo({required this.userId,
      required this.id, required this.title,
      required this.completed});
  @override
  List<Object?> get props => [userId, id, title, completed];
}
```

Equatable compares todos by their fields. Without Equatable, `==` compares identity: two separate instances are never equal.

Fetching todos with retry and backoff

1. Fetch `/todos` from the API and map each item with `Todo.fromJson`.
2. Load the app's todo list from this call instead of typing entries by hand.
3. Wrap the call in `withRetry` with a fixed maximum number of attempts.
4. Grow the delay exponentially between attempts.
5. Add random jitter so retries do not line up.
6. Point at an invalid path and log each attempt with its delay.

Done when

- [ ] Todos render from the live API on launch.
- [ ] An invalid path retries several times with growing delays.
- [ ] The retry log is on screen when you present.

The API

```
GET https://jsonplaceholder.typicode.com/todos
GET https://jsonplaceholder.typicode.com/todos/1
GET https://jsonplaceholder.typicode.com/nope
# One element:
{
  "userId": 1,
  "id": 1,
  "title": "delectus aut autem",
  "completed": false
}
```

Field names match Todo exactly. No renaming needed. `/nope` returns 404: the failure case for Task II.

Fetching the list

```dart
final _dio = Dio(BaseOptions(
  baseUrl: 'https://jsonplaceholder.typicode.com',
));
Future<List<Todo>> fetchTodos() async {
  final res = await _dio.get('/todos');
  final raw = res.data as List<dynamic>;
  return raw
      .map((j) => Todo.fromJson(j as Map<String, dynamic>))
      .toList();
}
```

JSON is decoded only in this function.

Backoff and jitter

```
Future<T> withRetry<T>(Future<T> Function() op) async {
  for (var i = 0; ; i++) {
    try {
      return await op();
    } catch (_) {
      if (i == 4) rethrow;          // five attempts, then give up
      final b = 400 * (1 << i);
      await Future.delayed(Duration(milliseconds: b + Random().nextInt(b)));
    }
  }
}
```

Retry only errors that can succeed later. A 404 fails on every attempt.

Errors and fixes

`JsonSerializable is undefined`	Put `json_annotation` in dependencies, not dev_dependencies.
`Undefined name _$TodoFromJson, or target of URI not generated`	Run `build_runner` from Setup; the `part` directive must name `todo.g.dart`.
`a == b is still false`	props is missing a field, or Todo does not extend Equatable.
`Undefined name Random`	Add `import 'dart:math';` to the file.

Presentation

Todos from the live API · the retry log on screen